# Lecture 2: CONVEX SETS and CONVEX FUNCTIONS

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# Outline

1. Convex sets
2. Convex functions

# Outline

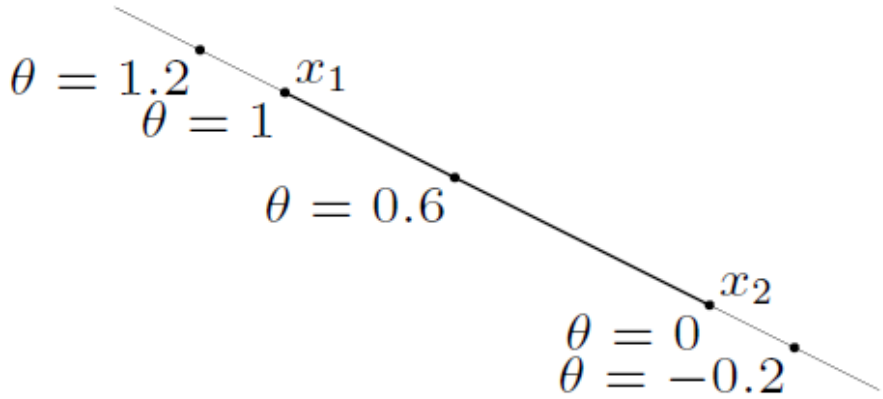
1 Convex sets

2 Convex functions

# Affine set

**line**: given $x_1, x_2 \in \mathbb{R}^n, x_1 \neq x_2$, a line through $x_1, x_2$ is

$$y = \theta x_1 + (1 - \theta)x_2, \forall \theta \in \mathbb{R}.$$




**affine set**: contains the *line* through any two distinct points in the set

# Convex set

- **line segment** between $x_1$ and $x_2$:

$$y = \theta x_1 + (1 - \theta)x_2, \forall \theta \in [0, 1]$$

- A set $C$ is a *convex set* if the line segment between any two points in $C$ lies in $C$.

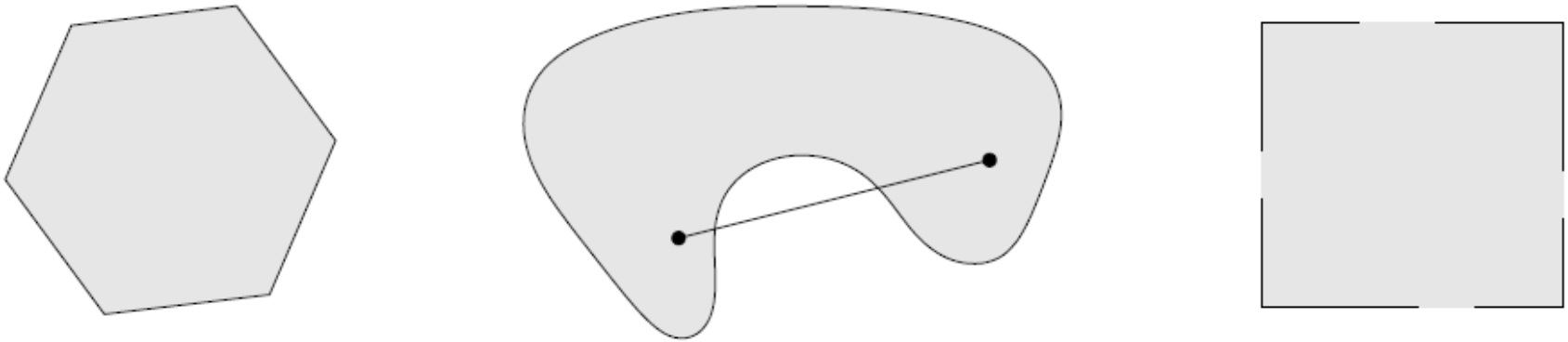


Figure: Left fig. is convex set, middle and right ones are non-convex sets.

Any affine set is convex set.

# Convex hull

The convex hull of a set $C$, denoted **conv**$C$, is the set of all convex combinations of points in $C$:

$$\textbf{conv}C = \{\theta_1 x_1 + \ldots + \theta_k x_k | x_i \in C, \theta_i \geq 0, i = 1, \ldots, k, \theta_1 + \ldots + \theta_k = 1\}$$

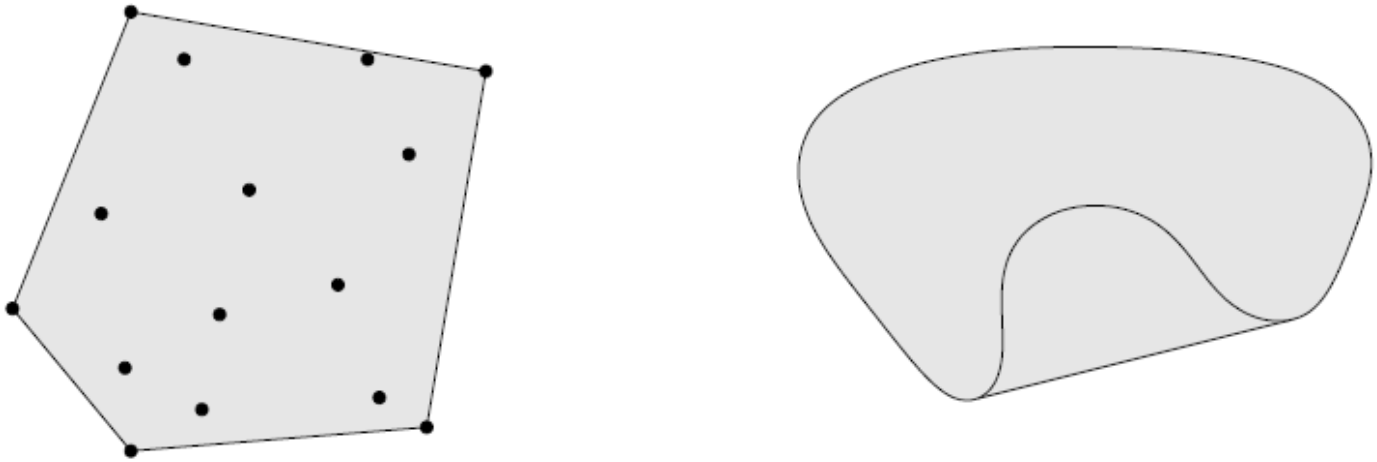


Figure: Some examples of convex hull.

Convex hull is always convex.

# Cones

A set $C$ is called a cone if for every $x \in C$ and $\theta \geq 0$, we have $\theta x \in C$.
A set $C$ is a *convex cone* if it is convex and a cone, i.e., for any $x_1, x_2 \in C$ and $\theta_1, \theta_2 \geq 0$, we have $\theta_1 x_1 + \theta_2 x_2 \in C$.

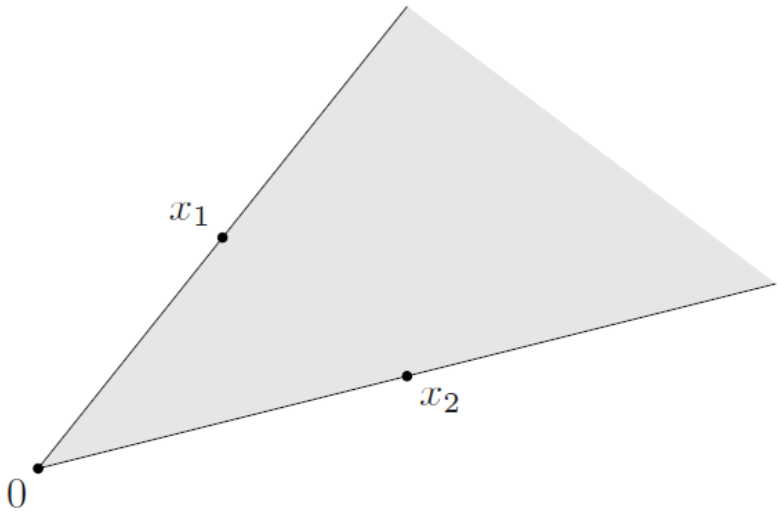



Figure: The pie slice shows all points of the form $\theta_1 x_1 + \theta_2 x_2$, where $\theta_1, \theta_2 \geq 0$.The apex of the slice corresponds to $\theta_1 = \theta_2 = 0$); its edges $\theta_1 = 0$ or $\theta_2 = 0$) pass through the points $x_1$ and $x_2$.

# Some examples

- Empty set $\emptyset$, any single point $\{x_0\}$, $\mathbf{R}^n$ are affine (hence, convex) sets.
- A line is a affine set. If the line passes through zero, it is a convex cone.
- A line segment is convex, but not affine.
- A ray is a convex cone, but not affine.
- $R^n_+$ is a convex cone.

# Properties

1. A set is convex if and only if its intersection with any line is convex.
2. A set is affine if and only if its intersection with any line is affine.
3. The convex hull of a set $S$ is the intersection of all convex sets that contain $S$.

# Hyperplanes

- A *hyperplane* is a set of the form $\{x \mid a^T x = b\}$, where $a \in \mathbf{R}^n, a \neq 0$ and $b \in \mathbf{R}$.
- Constant $b$ is the offset of the hyperplane from the origin.
- Another form of hyperplane: $\{x \mid a^T(x - x_0) = 0\}$.

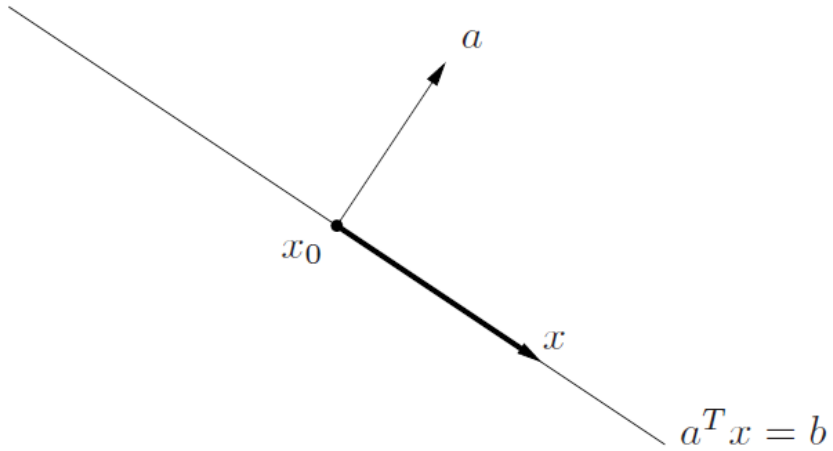



Figure: Hyperplane in $\mathbf{R}^2$, with normal vector $a$ and a point $x_0$ in the hyperplane. For any point $x$ in the hyperplane, $x - x_0$ (shown as the darker arrow) is orthogonal to $a$.

# Halfspaces

- A hyperplane divides $\mathbf{R}^n$ into two *halfspaces*. A (closed) halfspace is a set of the form $\{x|a^Tx \leq b\}$.
- A halfspace is convex, but not affine.
- Example of hyperplane in 2D:
  - The hyperplane $a^Tx - b = x_1 + 2x_2 - 2 = 0$ ($x_0 = (0, 1)$), normal vector $a = (1, 2)$, offset $b = 2$.
  - Halfspaces $x_1 + 2x_2 \geq 2$ and $x_1 + 2x_2 \leq 2$.

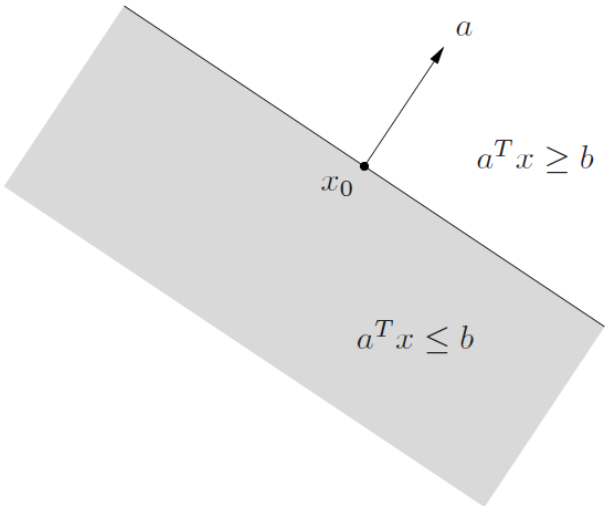



Figure: A hyperplane determines two halfspaces.

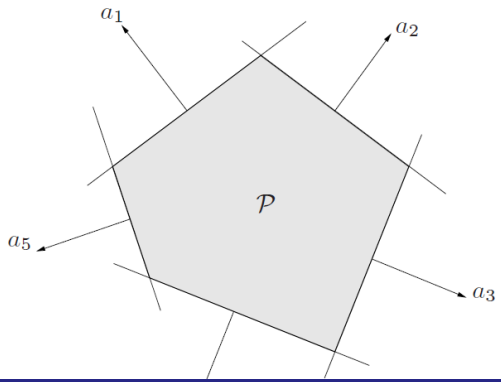
# Polyhedron

- A *polyhedron* is defined as the solution set of a finite number of linear equalities and inequalities. It is the intersection of a finite number of halfspaces and hyperplanes.

$$\mathcal{P} = \{x | a_j^T x \leq b_j, j = 1, \ldots m, c_j^T x = d_j, j = 1, \ldots p\}, or$$
$$\mathcal{P} = \{x | Ax \geq b, Cx = d\}.$$

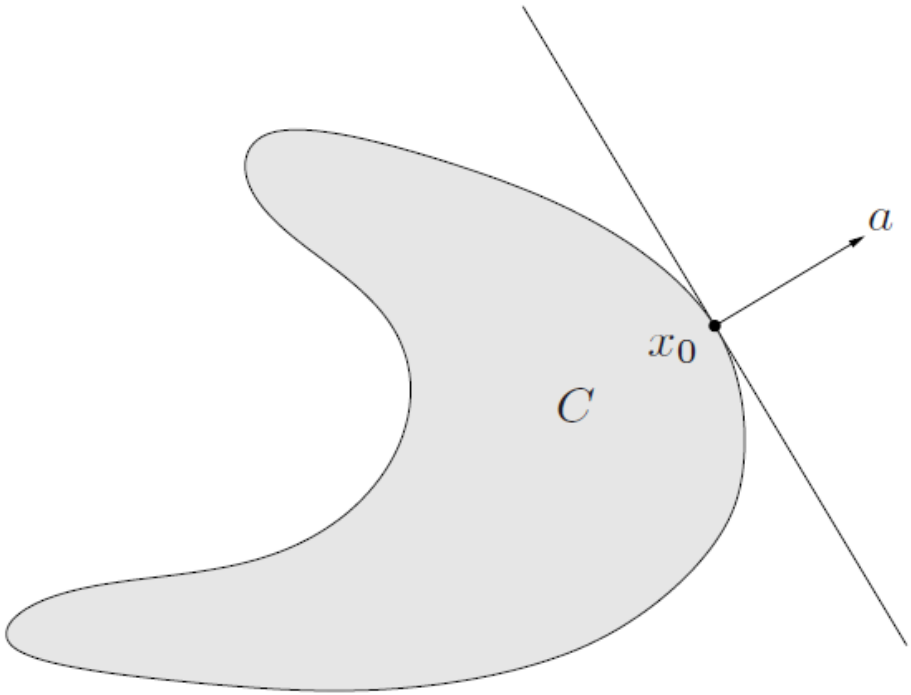
where $A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$, $C = \begin{bmatrix} c_1^T \\ \vdots \\ c_p^T \end{bmatrix}$, and $d = \begin{bmatrix} d_1 \\ \vdots \\ d_p \end{bmatrix}$.

- A polyhedron is a convex set.
- A bounded polyhedron is sometimes called a *polytope*.



# Supporting hyperplane

- If $a \neq 0$ satisfies $a^T x \leq a^T x_0$ for all $x \in C$, then the hyperplane $\{x | a^T x = a^T x_0\}$ is called a supporting hyperplane to $C$ at the point $x_0$.
- If a set is closed, has nonempty interior, and has a supporting hyperplane at every point in its boundary, then it is convex. (See exercise 2.27.)



# Operations that preserve convexity

- Intersection: the intersection of convex sets is convex.
- Scaling and translation: If $C$ is convex, then

$$aC + b = \{ax + b : x \in C\}$$

is convex for any $a, b \in \mathbb{R}$.

- Affine functions images and preimages: if $f(x) = Ax + B$ and $C$ is convex then

$$f(C) = \{f(x) : x \in C\}$$

is convex, and if $D$ is convex then

$$f^{-1}(D) = \{x : f(x) \in D\}$$

is convex.

# Operations that preserve convexity

- Perspective images and preimages: the perspective function is $P : \mathbb{R}^n \times \mathbb{R}_{++} \to \mathbb{R}^n$ (where $\mathbb{R}_{++}$ denotes positive reals),

$$P(x, z) = x/z$$

for $z > 0$. If $C \subseteq \mathrm{dom}(P)$ is convex then so is $P(C)$, and if $D$ is convex then so is $P^{-1}(D)$

- Linear-fractional images and preimages: the perspective map composed with an affine function,

$$f(x) = \frac{Ax + b}{c^T x + d}$$

is called a linear-fractional function, defined on $c^T x + d > 0$. If $C \subseteq \mathrm{dom}(f)$ is convex then so if $f(C)$, and if $D$ is convex then so is $f^{-1}(D)$

# Outline

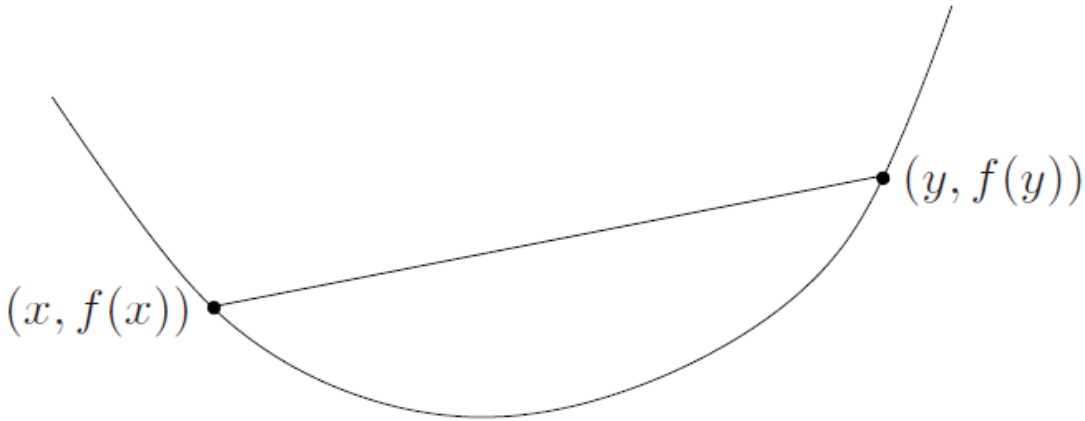
1 Convex sets

2 Convex functions

# Definition

A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if **dom** $f$ is a convex set and if for all $x, y \in$ **dom** $f$, and $\theta$ in $[0, 1]$, we have

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y).$$

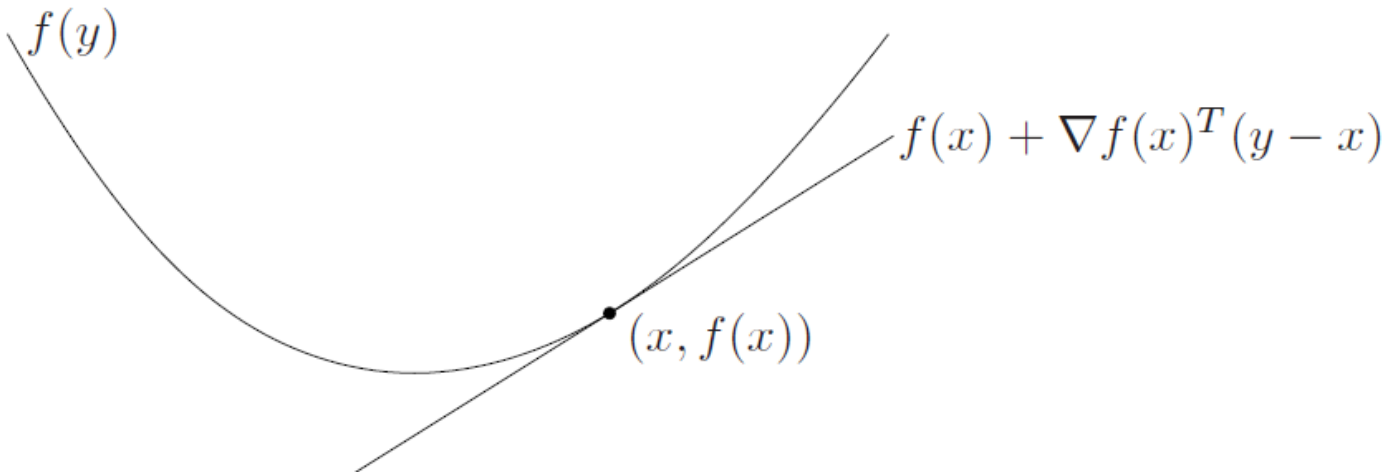



- Geometrically, the line segment between $(x, f(x))$ and $(y, f(y))$ lies above the graph of $f$
- $f$ is called *strictly convex* if the strict inequality holds
- $f$ is *concave* if $-f$ is convex, and *strictly concave* if $-f$ is strictly convex

# First-order conditions

Suppose $f$ is differentiable (i.e., its gradient $\nabla f$ exists at each point in $\mathrm{dom}\ f$, which is open). Then $f$ is convex if and only if **dom** $f$ is convex and

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \forall x, y \in \textbf{dom}\ f.$$



# First-order conditions (cont.)

$f$ is concave if and only if **dom** $f$ is convex **and**

$$f(y) \leq f(x) + \nabla f(x)^T (y - x), \forall x, y \in \text{dom } f$$

# Second-order conditions

Assume that $f$ is twice differentiable, i.e., Hessian , $\nabla^2 f$, exists at each point in **dom** $f$.

- $f$ is convex if and only if **dom** $f$ is convex and its Hessian is positive definite for all $x \in$ **dom** $f$ $(\nabla^2 f(x) \geq 0)$.
- $f$ is concave if and only if **dom** $f$ is convex and its Hessian is negative definite for all $x \in$ **dom** $f$ $(\nabla^2 f(x) \leq 0)$.

# Exercises

Prove that

- The quadratic function is convex: $f(x) = 10x_1^2 + x_2^2$.
- The entropy function is concave $f(x) = -\sum_i x_i \log(x_i)$.
- Ex. 3.1, page 113

# Examples

Quadratic function $f : \mathbb{R}^n \to \mathbb{R}$:

$$f(x) = (1/2)x^T Px + q^T x + r,$$

in which $P \in S^n, q \in \mathbb{R}^n, r \in \mathbb{R}$.

$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

$f$ convex if and only if $P \succeq 0$ (and concave if and only if $P \preceq 0$.)
Least-square objective $f(x) = \|Ax - b\|_2^2$

$$\nabla f(x) = 2A^T(Ax - b), \quad \nabla^2 f(x) = 2A^T A$$

convex for any $A$.

# Examples

- Quadratic-over-linear: $f(x, y) = x^2/y$, in which $x \in \mathbb{R}$ and $y \in \mathbb{R}, y > 0$, is convex.

$$\nabla^2 f(x) = \frac{2}{y^3} \begin{bmatrix} y \\ -x \end{bmatrix} \begin{bmatrix} y \\ -x \end{bmatrix}^T \geq 0$$

  convex for $y > 0$.

- Log-sum-exp: $f(x) = \log(e^{x_1} + \ldots + e^{x_n})$ is convex on $\mathbb{R}^n$

$$\nabla^2 f(x) = \frac{1}{1^T z} diag(z) - \frac{1}{(1^z)^2} zz^T \geq 0$$

  where $z_k = exp(x_k)$. (Please see the proof of $v^T \nabla^2 f(x) v \geq 0$ for any $v$ in book.)

- Geometric mean: $f(x) = (\prod_{i=1}^n x_i)^{1/n}$ is concave on $\mathbb{R}^n_{++}$

# Norms - recalled

**Norms** (denoted $\|x\|$) is a measure of length of a vector $x$. E.g., distance between two vector $x$ and $y$ is $\|x - y\|$.
A function $f : \mathbb{R}^n \to \mathbb{R}$ is called a **norm** if

- $f$ is nonnegative: $f(x) \geq 0, \forall x \in \mathbb{R}^n$
- $f$ is definite: $f(x) = 0$ only if $x = 0$
- $f$ is homogeneous: $f(tx) = |t|f(x)$ for all $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$
- $f$ satisfies: $f(x + y) \leq f(x) + f(y)$.

**!!! Every norm is convex !!!**

(Why?)

# Norms - examples

- Sum-absolute-value, or $\ell_1$-norm:

$$\|x\|_1 = |x_1| + |x_2| + \ldots + |x_n|.$$

- Chebyshev or $\ell_\infty$-norm:

$$\|x\|_\infty = \max\{|x_1|, |x_2|, \ldots, |x_n|\}.$$

- Euclidean norm, or $\ell_2$-norm:

$$\|x\|_2 = (x^T x)^{1/2} = (x_1^2 + x_2^2 + \ldots + x_n^2)^{1/2}.$$

- $\ell_p$-norm, $p > 1$:

$$\|x\|_p = (x_1^p + x_2^p + \ldots + x_n^p)^{1/p}.$$

- Quadratic norms. For $P$ is positive definite,

$$\|x\|_P = (x^T P x)^{1/2} = \|P^{1/2} x\|_2.$$

- Norms in $\mathbb{R}^{m \times n}$: Frobenius norm, maximum-absolute-value norm.

# Epigraph and sublevel set

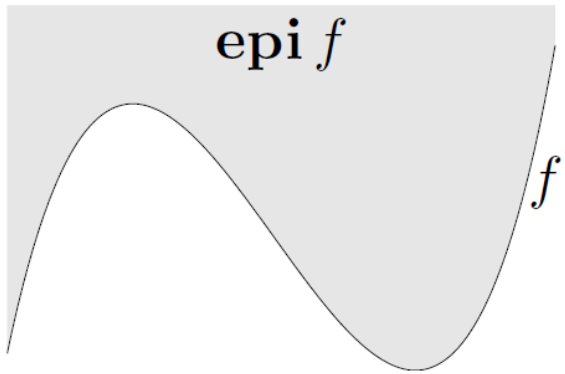
$\alpha$-**sublevel set** of $f : \mathbf{R}^n \to \mathbf{R}$:

$$C_\alpha = \{x \in \mathbf{dom}\, f \mid f(x) \leq \alpha\}$$

sublevel sets of convex functions are convex (converse is false)

**epigraph** of $f : \mathbf{R}^n \to \mathbf{R}$:

$$\mathbf{epi}\, f = \{(x,t) \in \mathbf{R}^{n+1} \mid x \in \mathbf{dom}\, f,\ f(x) \leq t\}$$




$f$ is convex if and only if $\mathbf{epi}\, f$ is a convex set

# Jensen's inequality and extensions

- If $f$ is convex, $x_1, \ldots, x_n \in \mathbf{dom} f$, and $\theta_1, \ldots, \theta_k \geq 0$ with $\theta_1 + \ldots + \theta_k = 1$, then

$$f(\theta_1 x_1 + \ldots + \theta_n x_n) \leq \theta_1 f(x_1) + \ldots + \theta_n f(x_n).$$

- The inequality extends to infinite sums, integrals, and expected values.
- If $p(x) \geq 0$ on $S \subset \mathbf{dom} f$, $\int_S p(x)dx = 1$, then

$$f\left(\int_S p(x)xdx\right) \leq \int_S f(x)p(x)dx.$$

- If $x$ is a random variable such that $x \in \mathbf{dom} f$ with probability 1, and $f$ is convex, then

$$f(\mathbb{E}x) \leq \mathbb{E}f(x).$$

# Other examples of convex functions (in $\mathbb{R}$)

- Linear and affine functions are convex (and concave).
- Exponential: $e^{ax}$ is convex on $\mathbb{R}$, for any $a \in \mathbb{R}$.
- Powers: $x^a$ is convex on $\mathbb{R}_{++}$ when $a \geq 1$ or $a \leq 0$, and concave for $0 \leq a \leq 1$.
- Powers of absolute value: $|x|^p$ is convex on $\mathbb{R}$ for $p \geq 1$.
- Logarithm: $\log(x)$ is concave on $\mathbb{R}_{++}$.

# Operations that preserve convexity

Nonnegative weighted sums of convex functions,

$$f(x) = w_1 f_1(x) + \ldots + w_m f_m(x),$$
$$g(x) = \int_A w(y) f(x,y) dy,$$

are convex.

Composition with an affine function Suppose $f : \mathbb{R}^n \to \mathbb{R}, A \in \mathbb{R}^{n\times} m, b \in \mathbb{R}^n$. Define $f : \mathbb{R}^m \to \mathbb{R}$, $g(x) = f(Ax + b)$, with $\mathbf{dom} g = \{x | Ax + b \in \mathbf{dom} f\}$. If $f$ is convex / concave, so is $g$.

- Log barrier for linear inequalities
  $f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x), dom(f) = \{x | a_i^T x < b_i, i = 1, \ldots, m\}$
- $f(x) = \|Ax - b\|$, where $A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m$, and $\|.\|$ is a norm in $\mathbb{R}^m$

# Operations that preserve convexity (cont.)

## Pointwise maximum

If $f_1$ and $f_2$ are convex functions, then pointwise maximum, $f(x) = \max\{f_1(x), f_2(x)\}$ with $\mathbf{dom} f = \mathbf{dom} f_1 \cap \mathbf{dom} f_2$ is also convex.

- $f(x) = \max_i \|A^{(i)}x - b^{(i)}\|$, where $A^{(i)} \in \mathbb{R}^{m\times n}$, $b^{(i)} \in \mathbb{R}^m$, and $\|.\|$ is a norm in $\mathbb{R}^m$ is a convex function
- Piecewise-linear functions $f(x) = \max\{a_1^T x + b_1, \ldots, a_n^T x + b_n\}$ is convex since it is the pointwise maximum of affine functions.
- Max function: $f(x) = \max\{x_1, \ldots, x_n\}$ is convex on $\mathbb{R}^n$.

# Operations that preserve convexity (cont.)

Composition. $g : \mathbb{R}^n \to \mathbb{R}, h : \mathbb{R} \to \mathbb{R}, f(x) = h(g(x))$.

- $f$ is convex if $h$ is convex and nondecreasing, and $g$ is convex,
- $f$ is convex if $h$ is convex and nonincreasing, and $g$ is concave,
- $f$ is concave if $h$ is concave and nondecreasing, and $g$ is concave,
- $f$ is concave if $h$ is concave and nonincreasing, and $g$ is convex.

Proof for $n = 1$, $f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$.
Examples:

- If $g$ is convex then $\exp g(x)$ is convex.
- If $g$ is concave and positive, then $\log g(x)$ is concave.
- If $g$ is concave and positive, then $1/g(x)$ is convex.
- If $g$ is convex and nonnegative and $p \geq 1$, then $g(x)^p$ is convex.
- If $g$ is convex then $-\log(-g(x))$ is convex on $\{x|g(x) < 0\}$.

# Examples of composition rules

$f(x) = -\log(-\log(\sum_{i=1}^{m} e^{a_i^T x + b_i}))$ on $\mathbf{dom}(f) = \{x | \sum_{i=1}^{m} e^{a_i^T x + b_i} < 1\}$ is convex since $\log(\sum_{i=1}^{n} e^{y_i})$ is convex.

# Operations that preserve convexity (cont.)

Minimization. If $f$ is convex in $(x, y)$, and $C$ is a convex set, then $g(x) = \min_{y \in C} f(x, y)$ is convex.

# Operations that preserve convexity (cont.)

Perspective of $f : \mathbb{R}^n \to \mathbb{R}$ is function $g : \mathbb{R}^{n+1} \to \mathbb{R}$ defined by

$$g(x, t) = tf(x/t), \forall t > 0.$$

If $f$ is a convex / concave function, then so is its perspective function $g$.
Examples:

- $f(x) = x^T x$ is convex, hence $g(x, t) = x^T x/t$ is convex for $t > 0$
- negative logarithm $f(x) = -\log(x)$ is convex, hence relative entropy $g(x, t) = t\log(t) - t\log(x)$ is convex.
- if $f$ is convex, then

$$g(x) = (c^T x + d) f((Ax + b)/(c^T x + d))$$

is convex on $\{x | c^T x + d > 0, (Ax + b)/(c^T x + d) \in dom(f)\}$

# Examples

- The entropy function $f(x) = -\sum_i x_i \log(x_i)$, where $x_i \in R,\ x_i > 0,\ \sum_i x_i = 1$ is concave.
- The relative entropy of two vectors $u, v \in \mathbb{R}^n$

$$\sum_{i=1}^{n} u_i \log(u_i/v_i)$$

is convex on $(u, v) \in \mathbb{R}^{2n}$.

**Proof.**

Consider the convex function $f(x) = -\log x, x \in R, x > 0$. Its perspective is

$$g(x,t) = -t\log(x/t) = t\log(t/x) = tlogt - t\log x,$$

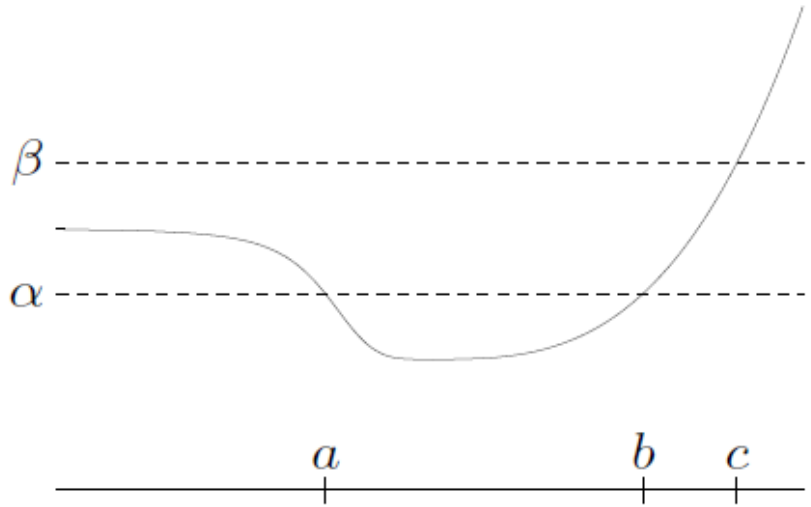
and is convex for all $x > 0$ and $t > 0$. The function $g$ is called the relative entropy of $t$ and $x$. For $x = 1$, $g$ reduces to the *negative entropy function*. The relative entropy of $u, v$ is convex since it is a sum of relative entropies of $u_i, v_i$. □

# Quasiconvex functions

$f : \mathbf{R}^n \to \mathbf{R}$ is quasiconvex if $\mathbf{dom}\, f$ is convex and the sublevel sets

$$S_\alpha = \{x \in \mathbf{dom}\, f \mid f(x) \leq \alpha\}$$

are convex for all $\alpha$




- $f$ is quasiconcave if $-f$ is quasiconvex
- $f$ is quasilinear if it is quasiconvex and quasiconcave

# Examples

- $\sqrt{|x|}$ is quasiconvex on $\mathbb{R}$
- $\log(x)$ is quasilinear on $\mathbb{R}_{++}$
- $f(x_1, x_2) = x_1 x_2$ is quasiconcave on $\mathbb{R}^2_{++}$
- Linear-fractional function is quasilinear

$$f(x) = \frac{a^T x + b}{c^T + d}, \quad dom(f) = \{x | c^T x + d > 0\}$$

# Log-concave and log-convex functions

A positive function $f$ is log-concave if $\log f$ is concave

$$f(\theta x + (1-\theta)y) \geq f(x)^{\theta} f(y)^{(1-\theta)}, \forall 0 \leq \theta \leq 1$$

$f$ is log-convex if $logf$ is convex.

- powers: $x^a$ on $\mathbb{R}_{++}$ is log-convex for $a \leq 0$, log-concave for $a \geq 0$
- sigmoid function
- many common probability densities are log-concave, e.g., normal:

$$f(x) = \frac{1}{\sqrt{(2\pi)^n det\Sigma}} exp(-\frac{1}{2}(x-\bar{x})^T \Sigma^{-1}(x-\bar{x}))$$

# Properties of log-concave functions

- twice differentiable $f$ with convex domain is log-concave if and only if

$$f(x)\nabla^2 f(x) \preceq \nabla f(x)\nabla f(x)^T.$$

- product of log-concave functions is log-concave
- sum of log-concave functions is not always log-concave
- integration: if $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is log-concave, then

$$g(x) = \int f(x, y)dy$$

is log-concave

# References

- Convex optimization book by Stephen Boyd
- Convex Optimization course at Stanford by Stephen Boyd
- Convex Optimization course at CMU by Ryan Tibshirani